



On the problem of slipper shapes of red blood cells in the microvasculature

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ABSTRACT

Red blood cells (RBC) are known to exhibit non symmetric (slipper) shapes in the microvasculature. Vesicles have been recently used as a model for RBC and numerical simulations proved the existence of slipper shapes under Poiseuille flow (both in unconfined and confined geometry). However, in our recent numerical simulations the transition from symmetric (parachute) shape to the slipper one was found to take place upon decreasing the flow strength, while experiments on RBCs showed the contrary. In this work we show that if the viscosity contrast (ratio between the internal over external fluid viscosities) is different from unity, as is the case with RBCs, the transition from parachute to slipper shape occurs upon increasing the flow strength, in agreement with experiments. We provide the phase diagram of shapes in the microcirculation. The slipper is found to have a higher speed than the parachute (for the same parameters), that we believe to be the basic reason for its prevailing in the microvasculature. We provide a simple geometrical picture that explains the slipper flow efficiency over the parachute one. Finally, we show that there exists in parameter space regions of co-existence of slipper/parachute shapes and suggest simple experimental protocols to test these findings. The coexistence of shapes seems to be supported by experiments, though a systematic experimental study is lacking. A potential application of this work is to guide designing flow-based experiments in order to link the shape of RBCs to pathologies affecting cell deformability, such as sickle cell diseases, malaria, and those affecting blood hematocrit, as in polycythemia vera disease.

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Introduction

Blood flow in the microvasculature corresponds to the highest dissipation (or resistance) site, according to the seminal work of the French physician [Poiseuille \(1828, 1830\)](#). This resistance depends both on the architecture of the microvascular network and on the rheological behavior of blood flowing through it, which, in turn, is governed by the shape of the RBCs. An understanding of the morphology of RBCs and their dynamics in the microvasculature is essential to understanding blood flow and blood diseases. Pathological alteration of RBCs and blood vessel deformability are known to be implicated in several diseases such as sickle cell diseases, malaria, microangiopathy and so on. A fascinating question and longstanding puzzle in the microvasculature is the fact that RBCs assume a non symmetric shape ([Skalak and Branemark, 1969](#)), called slipper.

Early studies modeling RBCs have been performed by [Secomb and Skalak, \(1982\)](#), and the slipper shape was later observed in numerical simulations by [Pozrikidis \(2005\)](#). These authors used a capsule as a model for RBC. Capsules are shells made of polymers and are endowed

with elastic properties, namely the shear elasticity that mimics the RBCs cytoskeleton (spectrin network lying underneath the cell membrane). The capsule, unlike RBC, has a stretching modulus. Vesicles (made of a bilayer of phospholipids) are inextensible (like RBC), but they are fluid, so that they are not endowed with membrane shear elasticity. However, in two dimensions, a situation in which we are interested, the notion of shear elasticity loses its meaning, and the vesicle model is equivalent to that of an RBC.

The identification of the minimal ingredients for the occurrence of a slipper shape was addressed more recently ([Kaoui et al., 2009](#)) where it has been shown that a two dimensional vesicle in an unbounded Poiseuille flow exhibits a slipper solution when the flow strength is comparable to that in the microvasculature, a slipper solution. The slipper solution was found to occur because the symmetric solution (called also parachute) becomes unstable. This instability corresponds to a bifurcation, called a pitchfork bifurcation, belonging to the so-called cusp catastrophe in the language of the catastrophe theory. This study has highlighted the fact that neither three dimensionality, nor confinements (mimicking blood vessels) are necessary ingredients for the manifestation of the slipper shape. Two outcomes of this study are: (i) the slipper moves faster than the parachute and (ii) the slipper is accompanied with membrane tank-treading (TT). It was speculated that TT mixes the internal RBC solution making available oxygen and ATP everywhere within the cell. This study was

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completed by introducing the lateral boundaries (Kaoui et al., 2011) mimicking blood vessels.

For ease of discussion it will be convenient to introduce now some notations. In the absence of RBCs, the plasma flow is of Poiseuille type and is given by

$$v_{0x} = v_{\max} \left[1 - \frac{y^2}{\left(\frac{w}{2}\right)^2} \right] \quad (1)$$

where w is the channel width, and the delimiting boundaries are located at $y = \pm w/2$, so that, $v_{0x}(y = \pm w/2) = 0$, and $v_{\max}$ is the maximum velocity. The shear rate at the wall is given by $\dot{\gamma}_w = 4v_{\max}/w$. The present model is characterized by four dimensionless parameters:

- The Capillary number measuring the flow strength over the bending energy of the membrane (where κ is the bending modulus, having a dimension of an energy; it is estimated for RBCs to 3×10^{-19} J. (Betz et al., 2009))

$$C_\kappa = \frac{\eta R^3 \dot{\gamma}_w}{\kappa} \quad (2)$$

- The degree of confinement

$$C_n = \frac{2R}{w} \quad (3)$$

where R is a typical radius of the vesicle and is defined as $R = \sqrt{A/\pi}$ where A is the enclosed area of the vesicle.

- The viscosity ratio measuring the ratio between the viscosity of the internal fluid, η' , to that of the external one η

$$\lambda = \frac{\eta'}{\eta} \quad (4)$$

- The reduced area (corresponding to the reduced volume in 3D)

$$v = \frac{A}{\pi \left(\frac{L}{2\pi}\right)^2} \quad (5)$$

which is the ratio of the actual area enclosed by the vesicles over that of a circle having the same perimeter, L , as the vesicle.

Our recent study (Kaoui et al., 2011) has shown that if the confinement is weak (small C_n), then the slipper solution obtained without the boundaries (Kaoui et al., 2009) was preserved. However, if the confinement is large enough, new branches of solutions took place leading to a complex phase diagram of shapes and dynamics, as summarized in Fig. 2. This phase diagram shows that for a given confinement C_n , the transition from the parachute shape to a slipper one always occurs upon decreasing the flow strength (which corresponds to decreasing C_κ). Experiments are sparse, but there is a consensus that the occurrence of the slipper happens upon decreasing the flow strength (Abkarian et al., 2008; Guido and Tomaiuolo, 2009; Tomaiuolo et al., 2009), which is in contradiction with our previous numerical study summarized in Fig. 2.

We have attempted to understand this major discrepancy, and have identified that the viscosity contrast plays a major role. RBCs in vivo have a viscosity contrast of about $\lambda \approx 5$, and this ratio depends on temperature. We have found that introduction of a viscosity contrast leads to new branches of slipper solutions that corresponds to a transition from parachute to slipper upon increasing the flow strength, in accord with the experiments. We also report here that there are new regions triggered by the viscosity contrast where there is coexistence between slipper and parachute solutions for the same parameters. We will provide the corresponding phase diagram

and attempt to make a quantitative analysis regarding the region of occurrence of slippers. A potential application of this work is the conception of new flow-based protocols and diagnosis in order to identify the link between RBCs shapes and blood pathologies such as sickle cell disease and polycythemia vera (see Conclusion).

Model and method

The studied system is schematically shown in Fig. 1. A two dimensional vesicle is immersed in a newtonian fluid bounded by two planar boundaries. The imposed flow is given by Eq. (1). At the scale of the RBC the Reynolds number is often small enough (exceptions may occur in the arterioles and at sites of aneurysms, for example, (Laadhari et al., 2012)) so that inertia can be neglected. The vesicle membrane acts on the fluid via bending forces, while the local membrane area remains unstretched; the membrane develops tension forces ζ that prevent extension (or compression). Both effects lead to the following membrane force per unit area (Kaoui et al., 2008):

$$\mathbf{f} = \kappa \left[\frac{d^2 c}{ds^2} + \frac{1}{2} c^3 \right] \mathbf{n} - \zeta \mathbf{c} \mathbf{n} + \frac{d\zeta}{ds} \mathbf{t} \quad (6)$$

Here c is the membrane curvature, $\mathbf{n}$ and $\mathbf{t}$ are the unit normal and tangent vectors on the membrane, and s is the arc length coordinate. We have extended our recent modeling without viscosity contrast (Kaoui et al., 2011) to the case with viscosity contrast. The Stokes equations can be converted into a boundary integral formulation that takes the form

$$\begin{aligned} \frac{\lambda + 1}{2} v(\mathbf{x}_0) &= v_{0x} + \frac{1}{\eta} \oint_{\text{vesicle}} \mathbf{G}(\mathbf{x} - \mathbf{x}_0) \cdot \mathbf{f}(\mathbf{x}) ds \\ &+ (1 - \lambda) \oint_{\text{vesicle}} v(\mathbf{x}) \cdot \mathbf{K}(\mathbf{x} - \mathbf{x}_0) \cdot \mathbf{n}(\mathbf{x}) ds \\ &+ \frac{1}{\eta} \oint_{\text{walls}} \mathbf{G}(\mathbf{x} - \mathbf{x}_0) \cdot \mathbf{f}_w(\mathbf{x}) ds \end{aligned} \quad (7)$$

$\mathbf{G}$ and $\mathbf{K}$ are second and third order tensors given by

$$\begin{aligned} G_{ij}(\mathbf{x} - \mathbf{x}_0) &= \delta_{ij} \ln |\mathbf{x} - \mathbf{x}_0| + \frac{(\mathbf{x} - \mathbf{x}_0)_i (\mathbf{x} - \mathbf{x}_0)_j}{|\mathbf{x} - \mathbf{x}_0|^2} \\ K_{ijk} &= -4 \frac{(\mathbf{x} - \mathbf{x}_0)_i (\mathbf{x} - \mathbf{x}_0)_j (\mathbf{x} - \mathbf{x}_0)_k}{|\mathbf{x} - \mathbf{x}_0|^4} \end{aligned}$$

Eq. (7) states that the membrane velocity (left term) is a sum of (i) the imposed flow, (ii) the induced flow due to the membrane force, (iii) the induced flow due to the fact that the viscosity inside is different from that outside, and (iv) the induced flow due to the boundaries, that we consider as quasi-rigid, endowed with a large elastic constant. This treatment is found numerically faster than imposing absolute rigidity of the walls.

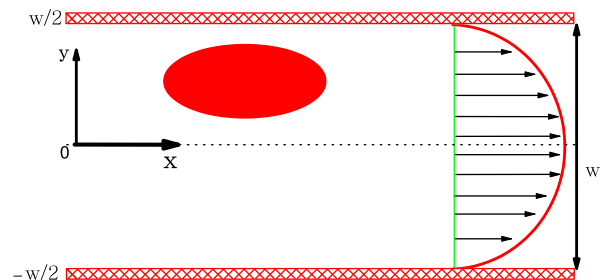


Fig. 1. A sketch of the system under consideration.

Results

We have set the viscosity contrast to $\lambda = 5$ (close to the physiological one for the human RBCs) (Cokelet and Meiselman, 1968). The reduced area is chosen to be $\nu = 0.6$.

The sites in the microvasculature where oxygen delivery occurs are arterioles (Fung, 1990) (with diameters of approximately 50 μm) and capillaries (with diameters from 5 to 10 μm); the corresponding average wall shear rates are about 8000/s and 1000/s, respectively. Membrane rigidity for a healthy cell is (Betz et al., 2009) $\kappa \approx 3 \cdot 10^{-19}$ J, and the typical RBC diameter (the diameter of a sphere having the same area) is about 6 μm . Finally, the plasma viscosity is $\eta \approx 10^{-3}$ Poise. Using these values, we found a range of $C_\kappa = 20$ –100. Greater membrane rigidity, as observed in infected cells (Suresh, 2006) (the rigidity of which may be one order of magnitude greater), yields smaller values for C_κ , with $C_\kappa \approx 1$ –10. We thus explored the range $C_\kappa = 1$ –200, which covers a wide interval of relevant values for healthy and unhealthy cells. The degrees of confinement of arterioles and capillaries are approximately 0.15 and 0.8, respectively. We explored the range of $C_n = 0.3$ –0.9 in our simulations.

We first concentrated on the regime of flow strength such that, $C_\kappa < 100$. In this case we have found that the phase diagram of shapes adopted by the vesicle (Fig. 3) is similar to the one obtained in the absence of viscosity contrast $\lambda = 1$ (Fig. 2). Note that we have explored here larger confinement values (as compared to Fig. 2) and we have discovered a new domain (red one) where there is a transition from OC-sn (off-centered snaking) to a steady elongated shape. We have checked that the same phenomenon occurs for $\lambda = 1$, not shown here. In other words, in this range of flow strengths, the overall feature of the phase diagram does not depend on the value of λ . However, there are some quantitative differences. Indeed, comparing Figs. 2 and 3, we observe that in Fig. 3 the phase diagram is shifted upward, to higher values of the capillary number C_κ . This is explained by the fact that for $\lambda = 5$ the effective rigidity (due to the increase of internal viscosity) of the vesicle is higher, and in order to, for example, trigger a parachute shape (starting from the snaking one), a higher flow is required to overcome the higher effective rigidity of the vesicle.

The interesting result with a viscosity contrast $\lambda = 5$ is that on further increase of the flow strength ($C_\kappa > 100$), the symmetric (parachute-like) regime becomes unstable in favor of slipper shape, as reported in experiments (Tomaiuolo et al., 2009). We have first fixed the confinement to $C_n = 0.35$, and have systematically varied C_κ in order to seek for non

symmetric solutions (slipper). Fig. 4 shows the lateral y -position of the center of mass in the channel, denoted as Y_G , as a function of C_κ . A symmetric shape has $Y_G = 0$, while a non symmetric shape is characterized by $Y_G \neq 0$. At small enough C_κ the shape is non symmetric (as is the case also for $\lambda = 1$); this is the slipper solution corresponding to the gray region in Fig. 4. Upon increasing the flow strength (increase of C_κ) the center of mass Y_G vanishes, meaning that the solution becomes symmetric (parachute). A novel behavior that emerges is that on increasing further the flow strength (increase of C_κ), the symmetric solution becomes unstable in favor of a non symmetric solution (slipper shape), as shown in Fig. 4. The same type of behavior in Fig. 4 has been found for various values of confinement C_n (we explored values in the range $C_n = 0.3$ –0.9).

The appearance of the slipper at large flow strength C_κ when the viscosity contrast is large enough, can be explained as follows. In the parachute regime, there is no tank-treading motion (due to symmetry). Thus the fluid inside behaves as a rigid solid. Increasing the flow strength, we observe that the parachute velocity lags further and further behind the maximal imposed velocity. This means that the velocity gradient around the vesicle becomes significant. Suppose, there is a slight shift upward of the center of mass (slipper solution), the slipper shape moves faster than the parachute, reducing thus the lag. However, the slipper is accompanied by membrane tank-treading, which induces dissipation inside the vesicle. The dissipation inside is weakly dependent on λ , since, due to stress balance at the membrane, we have as an order of magnitude, $\eta_{\text{in}} \nabla v_{\text{in}} \sim \eta_{\text{out}} \nabla v_{\text{out}}$. The last term is dominated by the imposed flow, making $\eta_{\text{in}} \nabla v_{\text{in}}$ to be fixed, basically, by the imposed flow. If the center of mass moves upwards (according to Fig. 1), the membrane makes counter-clockwise tank-treading, so that on the upper part it moves along the imposed flow (reducing the gradient), while in the lower plane, it moves against the flow, increasing the gradient. The birth of the slipper is in favor of a net gain in dissipation. Furthermore, for $\lambda = 5$, and in the weak confinement regime, the vesicle has a tendency to migrate outward, because its tank-treading angle become quite small (flow alignment). The wall induces a lift force that acts against this migration, so that in total the vesicle chooses an off-centered position due to the interplay of the lift force due to the flow curvature and that due to the wall. Further systematic studies are needed before completing our intuitive picture, however.

By repeating our exploration for different values of C_n and varying the flow strength, we have calculated the full phase diagram (Fig. 3).

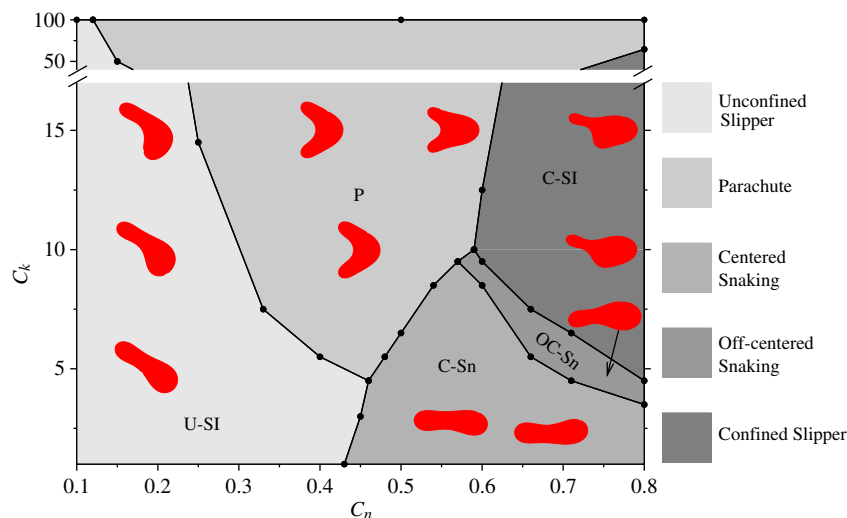


Fig. 2. The phase diagram of different shapes for $\lambda = 1$, as reported in Kaoui et al. (2011).

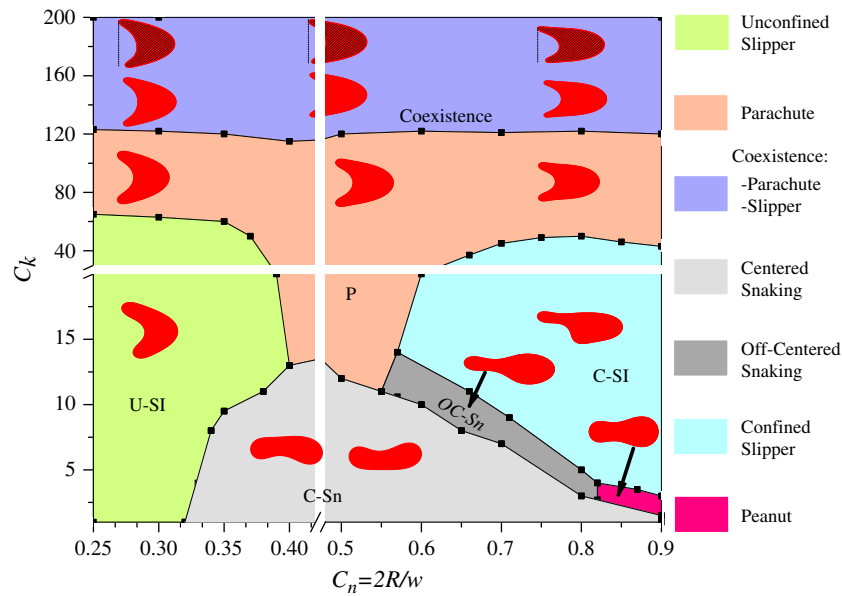


Fig. 3. The phase diagram for $\lambda=5$. In the range corresponding to about $C_\kappa < 120$, the overall picture of Fig. 2 is reproduced, with a new region (red one) corresponding to an elongated symmetric shape (called “peanut”). This type of solution is also found for $\lambda=1$. For $C_\kappa > 120$ we have a new domain where the parachute can transform into a slipper shape with possible coexistence between the parachute and slipper shapes. This new domain is not found for $\lambda=1$.

The slipper shape is shown with red color hatched with thin lines, in order to distinguish them from the parachute shape. In the absence of a viscosity contrast, we did not find any loss of stability of the parachute shape upon increasing C_κ (C_κ was increased up to 1000).

Actually, the situation turned out to be more subtle, since the slipper shape coexists with the parachute one. Whether one shape prevails over another depends on initial conditions. For example, if the initial shape is quite asymmetric, the tendency is in general (but not always) a convergence to a slipper. For the same parameters, if the initial shape is a centered ellipse then the final shape may either converge to a slipper or parachute depending on the parameters. This coexistence turns out to be present for all confinements explored so far. This coexistence seems to have been observed experimentally (see Discussion).

We believe that the reported experiments on RBCs (Abkarian et al., 2008; Guido and Tomaiuolo, 2009; Tomaiuolo et al., 2009) have always explored the high enough C_κ regime. Indeed they often observe a symmetric shape (which corresponds to the upper part of the orange domain in Fig. 3). Then upon increasing the flow strength (increasing C_κ)

the parachute transforms into a slipper. We shall come back more precisely to this point in the section devoted to discussion.

Discussion

We have seen that in the confined flow the slipper solution prevails at low enough and at large enough flow strength. The slipper is found to be faster than the parachute. A qualitative explanation is as follows. The slipper, despite having a center of mass outside the centerline, has its mass that is more confined around the centerline. This is seen on Fig. 5 where the parachute shape is compared to the slipper one. We have also computed the inertia moment with respect to the imposed flow centerline for each shape. This quantity informs us on how the representative points of the cell are distributed around the centerline. We find that the inertia moment of the slipper is smaller than that of the parachute shape (Fig. 6).

Experiments have reported (Abkarian et al., 2008; Tomaiuolo et al., 2009) that slipper flow takes place when increasing the flow strength. In the study by Abkarian et al. (2008) the capillary number is, in our definition, estimated to about 100 when their viscous pressure (their vertical axis in their Fig. 3 (b)) is equal to 1000 Pa. Their transition line from parachute to slipper is found experimentally equal to about 700 Pa, which would correspond in our units to about $C_\kappa=70$. Our numerical study shows a value that is about 120 (the

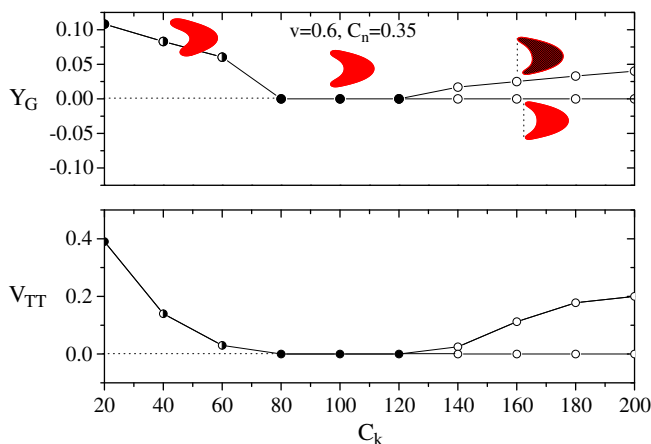


Fig. 4. The position of center of mass Y_G and the tank-treading velocity V_{TT} as a function of the flow strength.

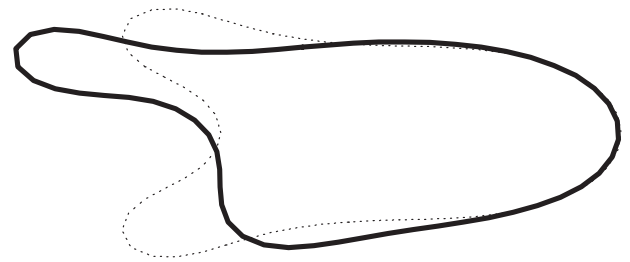


Fig. 5. The comparison of the slipper (full line) and the parachute (dotted line) shapes. The slipper, despite the fact that its center of mass is off-centered the total mass distribution is more focused around the flow centerline than the parachute shape ($C_n=0.8$ and $C_\kappa=20$).

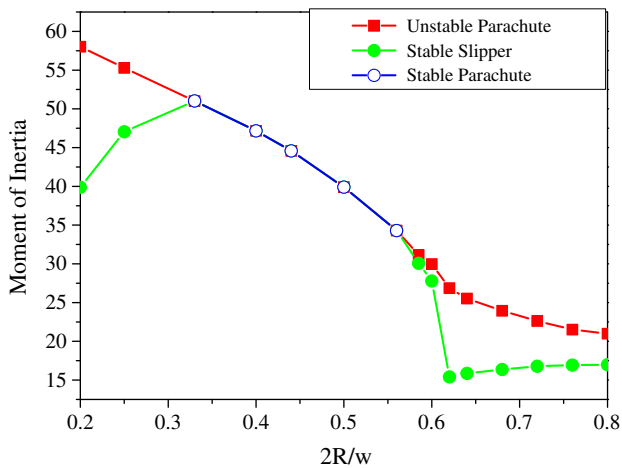


Fig. 6. The inertia moment for the slipper and parachute shape as a function of the confinement (for $C_k = 20$).

demarcation line between the orange and violet regions in Fig. 3). On one hand, given the uncertainties on the value of the membrane rigidity (where data vary from 2 to $3.5 \cdot 10^{-19}$ J), and on the other hand that our simulations are 2D, we find that the quantitative agreement is rather satisfactory. Note also that in the experimental report (Abkarian et al., 2008) the results of their Fig. 3b, in the slipper domain there are few shapes that seem rather a parachute, and this might be an indication of co-existence. In the experiments of Tomaiuolo et al. (2009) a slipper was observed for a velocity of about 3.6 cm/s, but not for a velocity 0.11 cm (see their Fig. 2). For a velocity of about 0.1 cm/s we have estimated $C_k = 10$, and in this case there is no slipper, in accord with our results. For the higher velocity we obtain $C_k = 350$, and in this case the RBC shows a slipper shape, in agreement with our finding (note that in our Fig. 3 we do not show higher values than 200, but we have checked that the slipper still exists for those values). In addition, for a velocity of 3.6 cm/s the experimental results show also another shape which is symmetric (not commented in the paper by Tomaiuolo et al. (2009) which seems also to support the idea of shapes coexistence).

It must be noted that an RBC is endowed with shear elasticity (due to cytoskeleton) with a shear modulus G . In principle, one should have also a capillary number associated with shear elasticity given by $C_s = \eta \dot{\gamma}_w R / G$. The quantity $\tau = \eta R / G$ is the typical time scale of relaxation of shape after cessation of the flow (if only shear elasticity were taken into account). The shear modulus is in the range $G = 0.006\text{--}0.009$ dyne/cm (Hochmuth et al., 1970; Waugh and Evans, 1979). Taking for $R = 3 \mu\text{m}$, one finds that τ is of about 0.5 ms. Shape recovery measurements (recovery of the biconcave shape after cessation of the flow) are estimated to about 0.1 s (Baskurt and Meiselman, 1996), which is more than two orders of magnitude larger than τ . Recently Tomaiuolo and Guido (2011) performed a start-up experiments shape dynamics of red blood cells in microcapillary flow, where they measured the time needed for the RBC to reach its terminal shape (steady shape under flow) after application of the flow. Their measured time, albeit interestingly twice larger than in recovery experiments (they found 0.2 s), still differs from the above introduced time τ by two orders of magnitudes. In order to resolve this dilemma it was suggested that there exists another time scale $\tau = \eta_m / G$, where η_m is the membrane viscosity, which is about 0.001 Poise cm. This leads to a time scale of about 0.1 s. This is of the order of the bending-based time scale. In two dimensions, the membrane is a line and therefore a shear mode does not make sense. Since our time scale is of about that observed for RBCs (0.1 s), we believe that the values of the capillary number defined by Eq. (2) should correspond to realistic

values for the RBCs. We should keep in mind that our model is simplistic and does not represent a full RBC model including membrane shear elasticity and viscosity. We hope, however, that the present study can serve as a guide for more elaborate studies.

Conclusion

The results reported in this work suggest several experimental protocols in order to link the shape and dynamics of flowing RBCs and pathologies affecting cell deformability. Malaria-infected cells exhibit an increase in the elastic membrane modulus. Exported proteins from parasites cause a significant increase in membrane rigidity, which may attain values that are 2.6 times higher than those of uninfected cells (Suresh, 2006). Similarly, in the case of hereditary hemolytic disorders such as spherocytosis, elliptocytosis, and sickle cell disease, the inherited mutations are found to markedly reduce the deformability of the RBC. Our phase diagram (Fig. 3) shows ample sensitivity of the RBC shape to membrane rigidity, that enters the definition of C_k . Thus from a benchmark of healthy RBC flow behavior to be used as a reference system, a systematic study of pathological conditions can be studied. For example, suppose that the confinement C_n and the flow strength are chosen for healthy RBCs in the parachute regime ("P" region in Fig. 3) close enough on top of the C-SI region of Fig. 3. For the same flow strength pathological cells, due to their stiffer membrane, would have smaller values of C_k and should fall in the C-SI domain instead of "P" regime. Thus capillary flow experiments are able to extract information on the pathological character by analyzing the adopted shapes. This kind of diagnosis should allow us to overcome the limits of classical static methods, such as micropipette aspiration, which are not suitable for handling a large number of cells.

Another possible application of these results is to link between RBCs shape and polycythemia vera disease in which bone marrow produces abnormally large hematocrit (Ht). The diagnosis would consist in studying blood flow in channels of diameters having several times the RBCs one. Due to a higher Ht each RBCs would feel locally higher confinement than in the case of normal Ht. Thus, for example if for normal Ht RBCs exhibit shapes in the "P" regime (by adequately choosing parameters; say about $C_k = 15$ with an effective confinement around 0.5), the majority of RBCs would exhibit a parachute shape. In contrast, for abnormal Ht the representative point in Fig. 3 would move to the right, potentially in the "confined-slipper" regime. A systematic analysis of RBCs shapes and dynamics under in vitro microcirculation should improve our comprehension of the behavior of RBCs in various pathologies, and ultimately help identifying the factors that impact thrombotic tendencies.

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References

- Abkarian, M., Faivre, M., Horton, R., Smistrup, K., Best-Popescu, C.A., Stone, H.A., 2008. Cellular-scale hydrodynamics. *Biomed. Mater.* 3, 034011.
- Baskurt, O., Meiselman, H., 1996. Determination of red blood cell shape recovery time constant in a Couette system by the analysis of light reflectance and ektacytometry. *Biorheology* 33, 489–503.
- Betz, T., Lenz, M., Joanny, J.F., Sykes, C., 2009. ATP-dependent mechanics of red blood cells. *PNAS* 106, 15320–15325.
- Cokelet, G., Meiselman, H., 1968. Rheological comparison of hemoglobin solutions and erythrocyte suspensions. *Science* 162, 275–277.
- Fung, Y., 1990. *Biomechanics: motion, flow, stress, and growth*. Biomechanics. Springer, New York.
- Guido, S., Tomaiuolo, G., 2009. Microconfined flow behavior of red blood cells in vitro. *C.R. Phys.* 10, 751–763.
- Hochmuth, R., et al., 1970. Capillary blood flow. I. Erythrocyte deformation in glass capillaries. *Microvasc. Res.* 2, 409–419.

- Kaoui, B., Ristow, G.H., Cantat, I., Misbah, C., Zimmermann, W., 2008. Lateral migration of a two-dimensional vesicle in unbounded Poiseuille flow. *Phys. Rev. E* 77, 021903.
- Kaoui, B., Biro, G., Misbah, C., 2009. Why do red blood cells have asymmetric shapes even in a symmetric flow? *Phys. Rev. Lett.* 103, 188101.
- Kaoui, B., Tahiri, N., Biben, T., Ez-Zahraoui, H., Benyoussef, A., Biro, G., Misbah, C., 2011. Complexity of vesicle microcirculation. *Phys. Rev. E* 84, 041906.
- Laadhari, A., Saramito, P., Misbah, C., 2012. Vesicle tumbling inhibited by inertia. *Phys. Fluid* 24, 031901.
- Poiseuille, J., 1828. D.Sc. thesis, Ecole Polytechnique Paris, 1828.
- Poiseuille, J., 1830. Recherches sur les causes du mouvement du sang sur les veines. *J. Physiol. Exp. Pathol.* 10, 277–295.
- Pozrikidis, C., 2005. Numerical simulation of cell motion in tube flow. *Ann. Biomed. Eng.* 33, 165–178.
- Secomb, T.W., Skalak, R., 1982. two-dimensional model for capillary flow of an asymmetric cell. *Microvasc. Res.* 24, 194–203.
- Skalak, R., Branemark, P.I., 1969. Deformation of red blood cells in capillaries. *Science* 164, 717–719.
- Suresh, S., 2006. Mechanical response of human red blood cells in health and disease: some structure–property–function relationships. *J. Mater. Res.* 21, 1871–1877.
- Tomaiuolo, G., Guido, S., 2011. Start-up shape dynamics of red blood cells in microcapillary flow. *Microvasc. Res.* 82, 35–45.
- Tomaiuolo, G., Simeone, M., Martinelli, V., Rotoli, B., Guido, S., 2009. Red blood cell deformability in microconfined shear flow. *Soft Matter* 5, 3736–3740.
- Waugh, R.E., Evans, E.A., 1979. Thermoelasticity of red blood cell membranes. *Biophys. J.* 26, 115–131.